Week 01: Research, Development and Innovation: Concepts and Technology Readiness

R&D and Project Management in Computer Science

R&D and Project Management in Computer Science (11117BLG002). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Funding bodies, tax authorities and statistical offices use precise definitions of research and development and of innovation. This week introduces the international definitions of the Frascati and Oslo manuals, the distinction between basic and use-inspired research and the technology readiness scale that many calls use [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to apply the five Frascati criteria to decide whether an activity is research and development. They should be able to distinguish basic research, applied research and experimental development, and to explain the Oslo Manual's definition of innovation. Finally, they should be able to assess the technology readiness level of a computing result.

Lecture

What counts as research and development

The Frascati Manual of the OECD defines research and experimental development as creative and systematic work undertaken to increase the stock of knowledge and to devise new applications of available knowledge [1]. An activity counts as R&D only if it meets five criteria, at least in principle: It must be novel, creative, uncertain in its outcome, systematic and transferable or reproducible. The manual distinguishes three types of R&D. Basic research seeks new knowledge without a particular application in view. Applied research seeks new knowledge directed towards a specific practical aim. Experimental development draws on knowledge and experience to produce new products or processes or to improve existing ones.

The criteria matter in computing, where much work consists of software development. Writing software is R&D when it resolves scientific or technological uncertainty on a systematic basis, and it is not R&D when it applies known techniques in routine ways. The migration of an application to a new framework version is normally routine. The design of a compiler optimisation whose effect cannot be predicted in advance may be experimental development.

What counts as innovation

The Oslo Manual, published by the OECD and Eurostat, defines an innovation as a new or improved product or process, or a combination of the two [2]. It must differ significantly from the unit's previous products or processes. It must also have been made available to potential users or brought into use by the unit. It distinguishes product innovations from business process innovations. Innovation therefore requires implementation, while R&D does not. A research project can produce knowledge without any innovation. An innovation can occur without any R&D, for example when a firm adopts a technology that is new to it.

Pasteur's quadrant

The familiar opposition between basic and applied research suggests a single line from theory to practice. Stokes argued that two separate questions are involved: whether research seeks fundamental understanding and whether it is guided by considerations of use [4]. The answers define quadrants named after exemplary scientists. Bohr's quadrant contains pure basic research, Edison's quadrant contains pure applied research, and Pasteur's quadrant contains use-inspired basic research, which seeks fundamental understanding while being motivated by a practical problem. Much research in computer science, from cryptography to machine learning, sits in Pasteur's quadrant, which helps explain why the same work can be presented to academic and industrial funders.

Technology readiness

Technology readiness levels were developed at NASA to assess the maturity of technologies, and Mankins reviewed their history and use [3]. The scale has nine levels, from basic principles observed at level 1 to an actual system proven in an operational environment at level 9, as Figure 1.1 shows. The European Union adopted the scale for its research programmes, where calls often state the level that a project should start from and the level it should reach. TÜBİTAK has also used the increase in technology readiness as an element of its evaluation of 1001 proposals [5]. For computing research, the difficult step is often the move from validation in the laboratory to validation in a relevant environment, which requires real data, real users or real infrastructure.

> ***Pause and reflect.*** *Place your current thesis or project on the technology readiness scale. What evidence would move it up one level?*

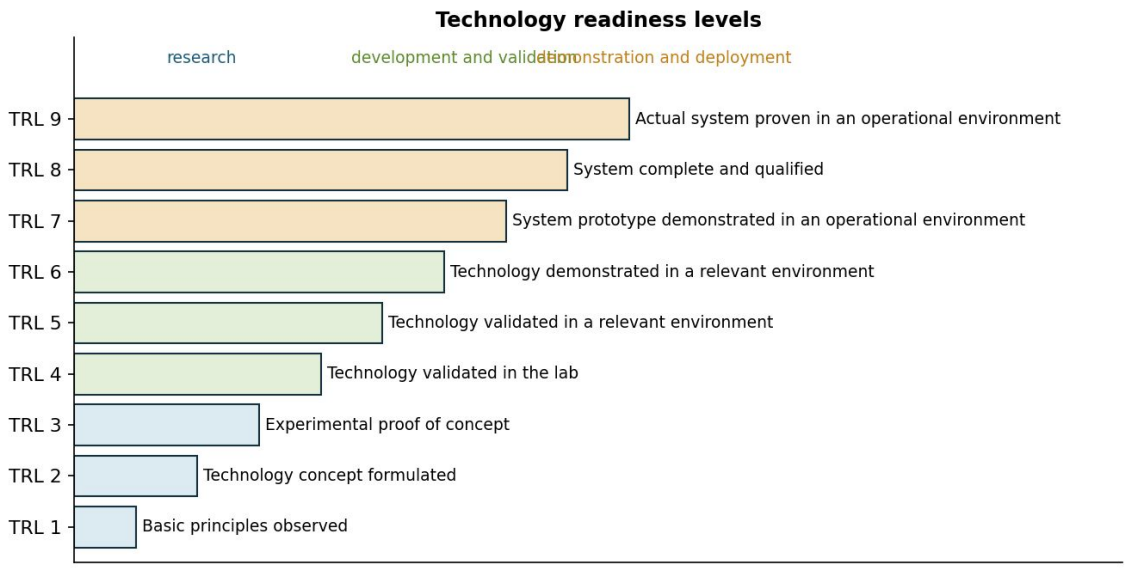


Figure 1.1. The nine technology readiness levels as used in European research programmes [3].

Hands-on components

Interactive lab: Part A applies the five Frascati criteria to computing activities. Part B sorts activities into the three types of R&D. Part C assesses technology readiness level by level [1, 3]. Link: https://utkukose.github.io/rd-project-management-NB-lecture/weeks/week-01/lab.html

Colab notebook: It encodes the five Frascati criteria as a function, classifies a portfolio of computing activities and summarises the R&D budget by type [1]. Link: https://colab.research.google.com/github/utkukose/rd-project-management-NB-lecture/blob/main/weeks/week-01/NB01_rd_innovation.ipynb

Reflection prompts

1. Which part of your current work would pass all five Frascati criteria, and which part would not? Why does the distinction matter for funding?
2. Where would you place your field in Stokes's quadrants, and how does that affect whom you ask for funding?
3. What evidence would reviewers expect before accepting a claimed technology readiness level in your area?

Weekly task and submission

What to do. Describe on one page a project idea that you will develop throughout the course. First, state the problem, the main question and the expected result in a few sentences. Second, list four to six activities of the project and check each against the five Frascati criteria. Third, place the project in one of Stokes's quadrants. Fourth, state its starting and target technology readiness levels, with the evidence that each level requires.

Where to look. Lecture: sections "What counts as research and development", "Pasteur's quadrant" and "Technology readiness". Interactive lab: Part A for the five Frascati criteria, Part B for the types of R&D and Part C for technology readiness. Colab notebook: Section 1 with Exercise 1, which encodes the Frascati criteria as a function.

Example. A project could detect cyberbullying in Turkish social media posts. Designing and testing a new detection model meets all five Frascati criteria, because its outcome is uncertain and the work is novel, creative, systematic and reproducible. Setting up a web server for a demonstration does not, because it is routine. The project seeks both understanding and use, which places it in Pasteur's quadrant. It could start at TRL 2 and aim for TRL 4, with the model validated in the laboratory on public data.

What to submit. The one-page description with a table of the activities and the criteria they meet, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] OECD (2015). Frascati Manual 2015: Guidelines for Collecting and Reporting Data on Research and Experimental Development. OECD Publishing. https://doi.org/10.1787/9789264239012-en

[2] OECD, & Eurostat (2018). Oslo Manual 2018: Guidelines for Collecting, Reporting and Using Data on Innovation (4th ed.). OECD Publishing. https://doi.org/10.1787/9789264304604-en

[3] Mankins, J. C. (2009). Technology readiness assessments: A retrospective. Acta Astronautica, 65(9-10), 1216-1223. https://doi.org/10.1016/j.actaastro.2009.03.058

[4] Stokes, D. E. (1997). Pasteur's Quadrant: Basic Science and Technological Innovation. Brookings Institution Press.

[5] TÜBİTAK (2020). ARDEB 1001 Programı: Proje değerlendirme sistemindeki yenilikler. https://tubitak.gov.tr/tr/duyuru/ardeb-1001-programi-proje-degerlendirme-sistemindeki-yenilikler